

Joint Augmented Reality Video Analytics and Artificial Intelligence Supervision

Benjamin Sho
b.sho@unsw.edu.au
CSIRO Data61
Sydney, Australia

Ryan A.J. de Belen
r.debelen@unsw.edu.au
UNSW
Sydney, Australia

Rowan T. Hughes
rowan.hughes@unsw.edu.au
UNSW
Sydney, Australia

Tomasz Bednarz
tomasz.bednarz@siggraph.org
CSIRO Data61 and UNSW
Sydney, Australia



Figure 1: Screen-captures from the app. (Left) Events are marked on the timeline with colors matching those of the corresponding event bounding-boxes visible in the video. (Right) Users can interact with the AI model in real-time, correcting classification errors and passing the information back to improve predictions.

ABSTRACT

Augmented Reality (AR) and Artificial Intelligence (AI) technologies have become ubiquitous in the public's daily lives. In the AR space, many notable industry players are engaged in bringing smaller, more ergonomic, and more powerful AR headsets and smart-glasses to the market, with the eventual goal of these devices to replace mobile phones for a wide variety of tasks. With the improvements in computing power on AR devices, a similar surge in AI-powered technologies running on these devices is also evident.

In this work, we present an AR application that explores Human-AI interaction in a computer vision context by providing an integrated platform for video analytics and AI model fine-tuning.

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1 INTRODUCTION

Human-AI interaction has been the subject of discussion for over two decades and is a highly active research area [Amershi et al.

2019]. The visualization of model output and the provision of training data are generally categorized as two distinct activities. AI practitioners often rely on large labeled datasets to train models, and these sets may not be optimal for a user's specific circumstances and purpose. There is often a gap in establishing a tailored feedback loop to correct erroneous outputs. One approach is to integrate the feedback loop into the same user interface used to visualize the output, enabling users to adapt models with their own data conveniently. This feedback loop bridges some of the gaps between Human-AI interactions.

Addressing the visualization of video analytics produced by Deep Learning (DL) models, an AR application that combines AI-powered video analytics with novel interaction design is presented. It allows users to record and/or load a video that is automatically annotated with the time, position and classification of events by a suitable DL model. Users can then visualize and interact with the event play-backs through hand gestures, voice commands and a holographic user interface. State-of-the-art DL models typically exhibit generalization error outside the domain of the model's training data [Jakubovitz et al. 2018]. One method of correcting AI misclassifications involves a technique known as transfer learning, which often requires labeled data in the target domain [Weiss et al. 2016]. The application provides capabilities for producing this data by integrating the AR device's in-built camera with a user interface, allowing the selection and relabelling of any misclassified events and establishing an intuitive human-AI feedback loop.

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2 VIDEO ANALYTICS AND USER INTERACTION

The application immerses users in an AR environment surrounded by a curved screen. Through a combination of hand gestures, body movements and voice commands, users can adjust the screen's position and size, providing up to 360° of viewing space. This ultra-wide viewing space enables the placement of complementary screens, in addition to the main screen, without the user feeling cluttered. The central screen consists of a rolling video, either pre-recorded or captured in real-time via the device's camera. The system uses a pre-trained model provided by the user to classify events of interest. Afterwards, it overlays bounding boxes that are colour-coded based on different events to draw user's attention to certain parts of the video. The complementary screen located to the left of the main screen showcases a previous event's snapshot. Concurrently, the screen to the right displays a snapshot of the next upcoming event.

The system supports multi-modal interaction to enhance human-AI collaboration. Users can use hand gestures to operate virtual toggles and buttons. Voice commands can also be used to control playback. During playback, users have access to a virtual timeline control (Figure 1 (Left)). In particular, users can pinch and drag the toggle to control video playback. The classified events are also presented on the timeline via markers with colour gradients corresponding to the detected events by the DL model, aiding user navigation. In addition, users can bring up a holographic menu to access a host of features through gesture input, including playback control, event filtering, event skipping, screen reset, video recording, toggling relabelling mode and more. Users can identify incorrectly classified events and relabel them accordingly (Figure 1 (Right)). The updated annotation data can subsequently be used for model retraining.

3 VIDEO ANNOTATION AND MODEL RETRAINING

Annotation metadata is read by the application in a JSON format, with a simple structure consisting of event position and begin/end frames. The application supports both offline and online model annotation. In the offline setting, a DL model's output can be pre-translated to JSON. Then this metadata and the accompanying video file can be streamed during run-time (Figure 2). For real-time classification and annotation, the Open Neural Network Exchange (ONNX) [Bai et al. 2019] format was used. Model output is then translated to a JSON format at run-time.

Often, single events appear as multiple events due to small gaps or frame misclassifications. These events can be merged using simple thresholding, tuned to user preference. This process can significantly improve user navigation between events on the timeline. To mitigate computational cost, particularly for longer videos, events are sorted by their frames, and a modified union-find data structure is used for merging them upon parsing the metadata. This allows multiple frames containing the same event to be merged in amortized constant time.

4 DISCUSSION & FUTURE WORK

We are currently amidst an explosion in the popularity of AI-powered enabled devices. These devices are presented with a tough

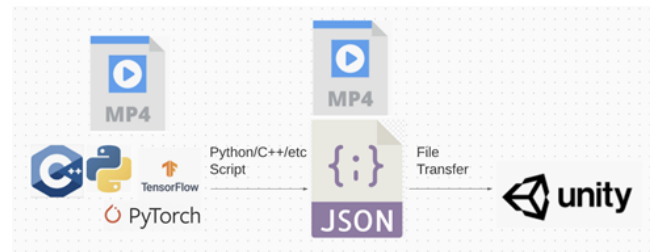


Figure 2: System flow diagram for offline annotation. For runtime annotation, the model is ported to ONNX with JSON generated in real-time.

challenge in classifying and navigating the world around them [de Belen et al. 2019a]. Humans could help to mitigate the problem with the aid of AR devices. Humans could aid such systems in real-time, helping with labeling of objects or correcting mistakes. Such an interface may also assist in improving trust in Human-AI interactions.

An obvious extension to the current framework would be integration with cloud-based services to transmit relabelled data and allow for live redeployment of AI models. Finally, integration with existing High-End Visualisation System (HEVS) [Bourke and Bednarz 2019] to allow multi-model collaborative visualisation between multiple devices [de Belen et al. 2019b; Nguyen and Bednarz 2020] at the same time, or with participatory dashboards [Lock et al. 2020].

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